Buddy System

Comprehensive Project Report | **UNAI × µLearn Hackathon 2026**

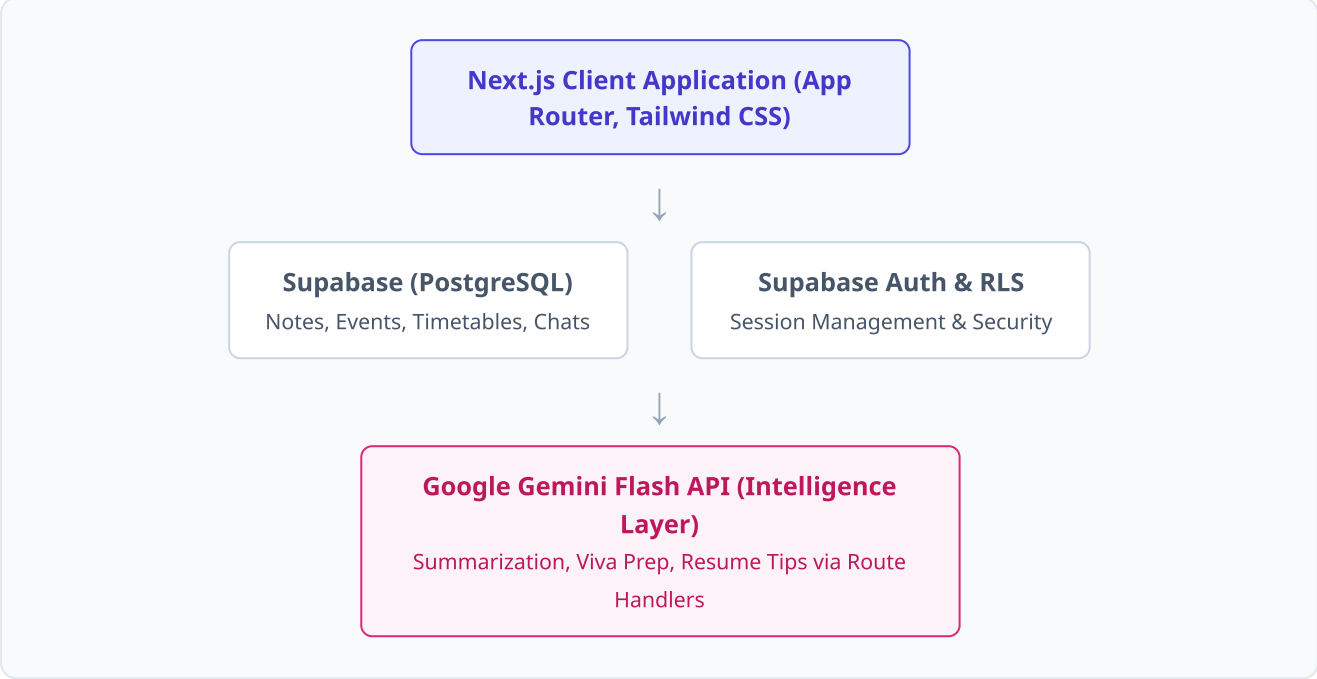
Team: **ERROR_404** | Repository: github.com/adarsh-naijo/Buddy_System

1. Executive Summary

The transition into engineering education presents severe logistical and social hurdles: fragmented class notes, isolated event announcements, uncoordinated exam revisions, and reluctance among freshmen to seek guidance from senior students. **Buddy System** solves this by establishing a unified campus platform. Built using Next.js, Supabase, and Google Gemini Flash API, the platform provides academic resource sharing, event discovery, personalized schedule tracking, and anonymous senior mentorship within an intuitive, high-performance interface.

2. System Architecture & Tech Stack

The application follows a modern serverless architecture, ensuring rapid edge-deployment and secure, real-time database syncing across the campus network.



Database Schema Contracts

- **notes:** Stores note metadata, semester, subject, module number, author, and cloud URLs.
- **events & event_rsvps:** Tracks upcoming technical workshops, dates, and dynamic user RSVP states.
- **timetables & exams:** Maintains period schedules, exam countdown timestamps, and user revision targets.
- **buddy_chat_rooms:** Powers real-time chat between anonymous freshers and seniors with alias preservation.

3. Key Modules & Functional Specifications

1. Note It (Resource Engine)

Centralized archive categorized by branch, semester, and module. Features direct integration with Gemini Flash to extract bulleted takeaways and generate oral viva examination questions from uploaded content.

2. Event Hub (Noticeboard)

A filterable showcase for technical symposiums, workshops, and club activities. Includes real-time RSVP counters, category filters, and dynamic registration link redirects.

3. Timetable & Exam Planner

Grid display of current day class schedule and room assignments. Integrated with exam countdown metrics and a community board for shared revision study plans.

4. Buddy Works (Mentorship)

Anonymous doubt broadcast board allowing juniors to ask foundational questions without fear of judgment. Supports live chat with identity masking and AI-driven resume tailoring.

4. Security & Privacy Safeguards

Protecting student data and encouraging honest communication without social anxiety are core design tenets of the Buddy System.

- **Row-Level Security (RLS):** Implemented at the PostgreSQL layer, ensuring users can only modify their own notes, RSVPs, and timetables.
- **Anonymity Layer:** Freshmen and mentor profiles in peer discussions use synthetic aliases (e.g., *Fresher_492*, *Senior_Mentor*). True identities are strictly hidden unless the senior mentor explicitly triggers a verified reveal.

5. Conclusion & Impact

Buddy System eliminates the communication friction prevalent in higher education institutions. By centralizing resources, automating study intelligence, and establishing a psychologically safe peer-learning network, the platform accelerates exam readiness and fosters authentic cross-batch collaboration.